Helia Vafaei

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LINKS

LinkedIn, GitHub, Personal website

EDUCATION

Sep 2020 – Present
B.Sc. in Computer Engineering, Iran University of Science and Technology Tehran, Iran
Among the top 4 universities in Iran according to QS University Rankings
GPA:
- Last two years: **3.58/4.00** (**17.56/20.00** on Iranian Scale)
- Overall: **3.58/4.00** (**17.27/20.00** on Iranian Scale)

Sep 2016 – Sep 2019
Diploma in Mathematics and Physics, Roshangaran High School Tehran, Iran
Overall Grade: **3.91/4.0** (**19.34/20** on Iranian Scale)

RESERCH INTEREST

Software Engineering
Software Development
Data Engineering
Data Science
Data Analytics
Software Verification and Testing
Programming Languages and Compilers

SKILLS

Programming Languages:
Python, React, HTML, CSS, JavaScript, C++, C#, C, SQL

Libraries and Frameworks:
React, Bootstrap, .Net, Keras, TensorFlow, Numpy

Environmental and Tools:
Git, Azure Devops, Linux, Bash, Postman

SELECTED PROJECTS

NoWaste:
- Participated as a Front-end developer in developing NoWaste, a web application aimed at minimizing food waste by connecting restaurants and cafes with surplus food to potential consumers.

- Designed and implemented RESTful APIs to facilitate efficient data exchange between frontend and backend systems.
- Dockerized the application to streamline deployment and ensure consistency across development and production environments.
- Developed and maintained CI/CD pipelines, automating testing, building, and deployment processes to enhance development efficiency and ensure robust application delivery.

Anti-spoofing algorithm:

- The project was for checking whether a face in the video is alive or not.
- In this project, two different models, including a feature-based and a deep learning model, were used.
- For the feature-based model, SVC was used.
- For the deep model, VGG16 and some extra un-pretrained layers were used.
- The data came from CASIA-FASD.

Computer system security project:

- The main aim of this project is retrieving information from the victim device using Bash in Linux and displaying it in a simple website.

Software quality measurement and its quantification:

- By running this project, the quality of a software can be measured quantitatively. In the software quality article, factors are presented that depend on a number of criteria, but the criteria is qualitative, so we use a number of metrics to achieve a quantity.

HONORS AND CERTIFICATES

Oct 2024	Ranked within **the Top 10% among about 100** in my class in GPA at IUST
Jul 2022	Ranked within **the Top 1% among about 155000** in Iranian University Entrance Exam
Oct 2023	Programming for Everybody (Getting Started with Python), Coursera
Jul 2022	Algorithms on Strings, Coursera
Mar 2022	Algorithms on Graphs, Coursera
Dec 2021	Data Structures, Coursera
Nov 2021	Algorithmic Toolbox, Coursera

ACADEMIC EXPERIENCE

Honorary Member

- Voluntary work in Computer Engineering Scientific Association at Iran University of Science and Technology (CESA IUST) – Fall 2023

Teaching Assistant

- Systems Analysis and Design – Instructor: Dr. Mehdad Ashtiani – Fall 2024
- Data Transmission – Instructor: Dr. Marzieh Sheikhi – Spring 2024
- Artificial Intelligence – Instructor: Dr. Arash Abdi – Fall 2023
- Database Design – Instructor: Dr. Hossein Rahmani – Spring 2023
- Fundamentals of Computer Programming – Instructor: Dr. Marzieh Davoodabadi Farahani – Fall 2022
- Advanced Computer Programming – Instructor: Dr. Sauleh Eetemadi – Spring 2022
- Advanced Computer Programming – Instructor: Dr. Sauleh Eetemadi – Fall 2021

WORK EXPERIENCE

Jul 2024 – present **Teacher, Farzanegan High School**

- Farzangan High School is one of the high schools of the National Organization for the Development of Talents (SAMPAD).
- I have been working as programing computer teacher.

Aug 2023 – Sep 2023 **Front-end Developer, Megamouj-e Iranian**

- Iranian Megamoj Company is a first-class manufacturing and industrial knowledge-based company in the field of telecommunications and computers.
- My main responsibilities included designing and developing various sections of the Eskab (School Service System) website.

LANGUAGES

Persian: Native Proficiency
English : Full Professional Proficiency IELTS (Academic): 7.0 (Overall score)

REFERENCES

Dr. Sauleh Eetemadi:

- Assistant Professor in the Department of Computer Engineering at IUST
- sauleh@iust.ac.ir

Dr. Mehrdad Ashtiani:

- Assistant Professor in the Department of Computer Engineering at IUST
- m_ashtiani@iust.ac.ir

Dr. Marzieh Malekimajd:

- Assistant Professor in the Department of Computer Engineering at IUST
- malekimajd@iust.ac.ir